Practical No.8

Que :- A system has RAM which can store four pages simultaneously to satisfy page requests. Write a Program to calculate percentage page hit and page fault if the system uses Least recently used page replacement technique for the page references entered by the user i.e. 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5 (No of Frames=3)

Code:

```
#include <stdio.h>

int main() {
   int pages[] = {1,2,3,4,1,2,5,1,2,3,4,5};
   int n = 12;
   int frames[3], time[3];
   int i, j, k, pos, faults = 0, hits = 0;
   int counter = 0;

   // Initialize frames
   for(i = 0; i < 3; i++) {
      frames[i] = -1;
      time[i] = 0;
   }

   for(i = 0; i < n; i++) {
      int found = 0;

      // Check for hit
      for(j = 0; j < 3; j++) {
         if(frames[j] == pages[i]) {
            hits++;
            counter++;
            time[j] = counter;
```

```
            found = 1;
            break;
        }
    }

    // If not found → page fault
    if(!found) {
        pos = 0;

        // Find least recently used
        for(j = 1; j < 3; j++) {
            if(time[j] < time[pos]) {
                pos = j;
            }
        }

        frames[pos] = pages[i];
        counter++;
        time[pos] = counter;
        faults++;
    }
  }

  printf("Total Page Faults = %d\n", faults);
  printf("Total Page Hits = %d\n", hits);

  printf("Page Fault Percentage = %.2f%%\n", (float)faults/n*100);
  printf("Page Hit Percentage = %.2f%%\n", (float)hits/n*100);

  return 0;
}
```

```
hari_ai@Harryfunlapcare: ~
hari_ai@Harryfunlapcare:~$ nano lru.c
hari_ai@Harryfunlapcare:~$ gcc lru.c -o lru
hari_ai@Harryfunlapcare:~$ ./lru
Total Page Faults = 10
Total Page Hits = 2
Page Fault Percentage = 83.33%
Page Hit Percentage = 16.67%
hari_ai@Harryfunlapcare:~$
```